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Brain and Blood Biomarkers of Major Depressive Disorder
A Systematic Review

Lia J. Zallar, PhD; Maura B. Dupont, MD, PhD

IMPORTANCE Major depressive disorder (MDD) is a debilitating neuropsychiatric disease associated with high risk of death by suicide. MDD is heterogeneous, and genome-wide association studies (GWASs) have established that MDD is polygenic, reporting thousands of small-effect variants. There is a significant epigenetic component in MDD, reflecting the role of gene-environment interactions in its pathogenesis. No biomarkers or precision-medicine based treatments exist. Understanding brain methylome and gene expression profiles at the regional and cellular level and whether brain molecular pathways alterations are reflected in blood could help identify phenotype-specific biomarkers and treatment targets for MDD.

OBJECTIVES To integrate and reanalyze findings from bulk and single-nuclei omics of MDD and to determine whether they reveal overlapping brain and blood dysregulated pathways that may inform biomarker and therapeutic targets discovery.

EVIDENCE REVIEW Omic studies in MDD brain and blood were reviewed using PubMed, Embase, and Web of Science. Articles were included if they used human biospecimens, reported differentially expressed or methylated genes when comparing MDD with control, and used RNA sequencing or methylation profiling. In addition, bulk and single nucleus differentially expressed genes were compared with the NHGRI-EBI GWAS Catalog of MDD-associated gene variants.

FINDINGS In the 54 articles included (30 in brain, 20 in blood, and 4 in both), 744 differentially expressed genes were altered in the same direction in brain and blood across studies, 43 of which overlapped with GWAS-identified MDD risk loci. A total of 544 differentially methylated genes were altered in the same direction in brain and blood across studies, 34 of which overlapped with GWAS-identified MDD risk loci. Several hub genes converged on developmental, inflammatory, transcriptional, apoptotic, and mitochondrial pathways.

CONCLUSIONS AND RELEVANCE While the individual dysregulated genes are highly variable across studies and brain regions, converging pathways and hub gene functions, including those related to neurodevelopment, mitochondria, neuroinflammation, apoptosis, and transcriptional regulation emerge across studies. Cell-type specific multiomic studies involving understudied brain regions are needed to better understand the pathophysiology of MDD and identify relevant biomarkers and treatment targets.

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Author Affiliations: Department of Psychiatry, Columbia University, New York, New York (Zallar, Dupont); New York State Psychiatric Institute, New York (Zallar, Dupont).

Corresponding Author: Maura B. Dupont, MD, PhD, Department of Psychiatry, Columbia University, 1051 Riverside Dr, Unit 42, New York, NY 10032 (mb928@columbia.edu).

Major depressive disorder (MDD) will be the leading cause of disability by 2030[1] due to chronicity, treatment resistance, and suicide rates. To reduce suffering and costs, molecular biomarkers and therapeutic targets are needed.

MDD cognitive, emotional, psychomotor, and neurovegetative symptoms reflect dysfunction across multiple cortico-limbic regions.[2] The dorsolateral prefrontal cortex (dlPFC) controlling executive functions,[3] the subgenual anterior cingulate cortex (sgACC) involved in decision-making,[4] and the orbitofrontal cortex (OFC) regulating impulse control[5] have reduced volume and connectivity in MDD. The hippocampus, controlling memory and emotional responses, shows less connectivity with the PFC and amygdala[6] and a smaller dentate gyrus and ventral subiculum (vSUB).[7] Amygdala hyperactivation alters emotional processing,[8] and temporal pole dysfunction contributes to altered reappraisal.[9] Altered activity in the nucleus accumbens (NAc) affects the reward circuitry,[10] and dysregulated anterior insula (aINS) affects integration of sensory, emotional, cognitive, and interoceptive information.[11]

Joint analysis of genome-wide[12,13] and whole-methylome[14] studies is necessary for understanding genetic and epigenetic[15,16] mechanisms of MDD pathogenesis. To identify disease biomarkers and new treatment targets, in a systematic review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guideline, we compiled and reanalyzed datasets from brain and blood multiomic and genome-wide association studies (GWASs).

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Methods and Literature Search

We searched PubMed, Embase, and Web of Science for studies conducting transcriptomic or methylomic analysis in MDD, using the following terms: *major depressive disorder*, *RNA sequencing*, *gene expression*, *single nucleus*, *methylation*, *brain*, and *blood*. After removing duplicates, titles and abstracts were screened following the PRISMA workflow (eFigure 1 in Supplement 1). Without a time limit for when articles were published, we selected studies that (1) conducted RNA sequencing (RNA-seq) in bulk brain tissue, brain single nuclei (snRNA-seq), whole blood, or blood cells, and/or genome-wide methylomic analysis of brain or blood and (2) shared databases of reported differentially expressed (DEGs) or differentially methylated genes (DMGs) in MDD. We developed a quality assessment tool using criteria informed by our experience performing omics studies and best practices for RNA-seq and methylomics. First we evaluated whether the article shared gene lists. That was the strongest factor. Then we assessed the quality of the statistical analysis: whether the study reported which covariates they used in the analyses, whether they shared their statistical threshold, whether they had a method for correcting for multiple testing, whether they shared their log-fold change cutoff, and the overall differential analysis method. Then we checked the study sample size and if they reported findings by sex (eTable1 in Supplement 2). Compiled DEG and DMG lists across brain and blood were extracted from the 54 studies included. We used a Python script on Google Colab to query gene Application Programming Interfaces from Ensembl, UniProt, GeneCards, and NCBI Gene for all 22 897 genes found in bulk brain, and subsets found in whole blood, and specific brain and blood cell types, and we functionally annotated them using Ensembl BioMart (eTables 2-6 in Supplement 2). Then we compared genes across studies for direction of change in MDD brain and blood (eTables 7 and 8 in Supplement 2). Finally, to identify which DEGs and DMGs had gene variants associated with MDD, we integrated those gene lists with the NHGRI-EBI GWAS Catalog[17] (eTables 9 and 10 in Supplement 2).

We performed gene ontology enrichment analysis using clusterProfiler version 4.8.3 limited to biological process. Gene symbols were converted to Entrez IDs (org.Hs.eg.db) and results filtered at $P < .05$ and $q < 0.2$. Terms with fewer than 10 genes or no significant enrichment were excluded, and the top 10 terms per group were visualized.

Protein-protein interaction (PPI) networks were generated for the top 300 protein-coding DEGs in each brain region, brain cell type, and blood sample type, using STRING version 11.5 (confidence score threshold ≥200). Hub genes were those showing highest degree of centrality (number of direct connections per node; eTable 11 in Supplement 2). The top 10 hub genes were visualized using binary presence matrices and heatmaps (pheatmap) and annotated using gene ontology analysis for biological process (eTables 12-14 in Supplement 2).

Results

Retrieved and Included Articles

The systematic literature search yielded 7325 results, with 2743 duplicates. After title and abstract review, 167 studies underwent full text evaluation. Fifty-four studies met the criteria (performing transcriptomics or methylomics in brain or blood and sharing gene lists) and were included (eFigure 1 in Supplement 1).

Key Points

Question What are the available multiomic data on major depressive disorder (MDD) genetic and epigenetic dysregulation and what molecular pathways are involved in disease pathogenesis?

Findings This systematic review found differentially expressed and differentially methylated genes in MDD brain and blood that also have variants that have been associated with MDD in GWAS databases.

Meaning The findings suggest the possibility of dysregulated molecular pathways in MDD that are detectable in the blood and can provide biomarkers for patient phenotypical stratification and treatment targets.

Study Characteristics

The 54 eligible publications, years 2014 to 2025 (eTable 1 in Supplement 2), included 17 065 participants, 48.5% women (eTable 1 in Supplement 2; eFigure 1 in Supplement 1). Thirty profiled brain only (16 bulk RNA-seq; 4 snRNA-seq; 2 both, but 1 reported only bulk DEGs[18]; 5 methylomics; 2 bulk RNA-seq and methylomics; 1 bulk RNA-seq, snRNA-seq and methylomics without listing snRNA-seq DEGs[19]). Twenty profiled blood only (9 RNA-seq, 9 methylomics, 2 both), and 4 studied both brain and blood (3 methylomics, 1 blood RNA-seq and brain snRNA-seq). Among the 54 studies, 29 originated from the US, 12 from Canada, the rest from the United Kingdom (7), Korea (2), China (2), Spain (1), France (1), Germany (1), and Israel (1), and some involved international collaborations (US-Canada, USA-United Kingdom-Canada, Canada-Taiwan, UK-Australia, and US-Netherlands). Thirty-nine studies did not report participants' ancestry breakdown, and those reporting were heavily dominated by White or European samples (79.9%), followed by Asian (9.6%), African (6.8%), and other (3.2%).

Bulk Transcriptomics

Brain Regions

We obtained bulk brain DEGs from 21 RNA-seq studies (2018-2024) on the dlPFC,[18-28] rostral ACC,[29] sgACC,[19-23,25,30-32] OFC,[19-23,25,28,33] aINS,[19-23,25] NAc,[19-23,25] temporal pole,[34] amygdala (midbody,[35] unspecified subregion[31]), hippocampus vSUB[19-23,25] and dentate gyrus,[36] lateral habenula,[37] and pooled brain samples.[38]

Gene ontology analysis on all 20 847 DEGs[18-38] (eTable 2 in Supplement 2) showed upregulated pathways controlling metabolism, translation, and mitochondria across dlPFC, sgACC, OFC, and aINS (**Figure 1**). Region-specific upregulated pathways were RNA splicing in aINS; gliogenesis, neurogenesis, and cellular development in NAc; and antigen processing in vSUB. Pathways involved in brain development, cell-cell adhesion, and axon guidance were downregulated across dlPFC, sgACC, aINS, NAc (**Figure 2**). Region-specific downregulated pathways were RNA splicing in OFC and amygdala (eFigure 3 in Supplement 1); DNA repair in amygdala; synaptic organization, structure, assembly, plasticity, and learning and memory in vSUB; and T cell and leucocyte differentiation in aINS. Other brain regions did not yield significant gene ontology terms.

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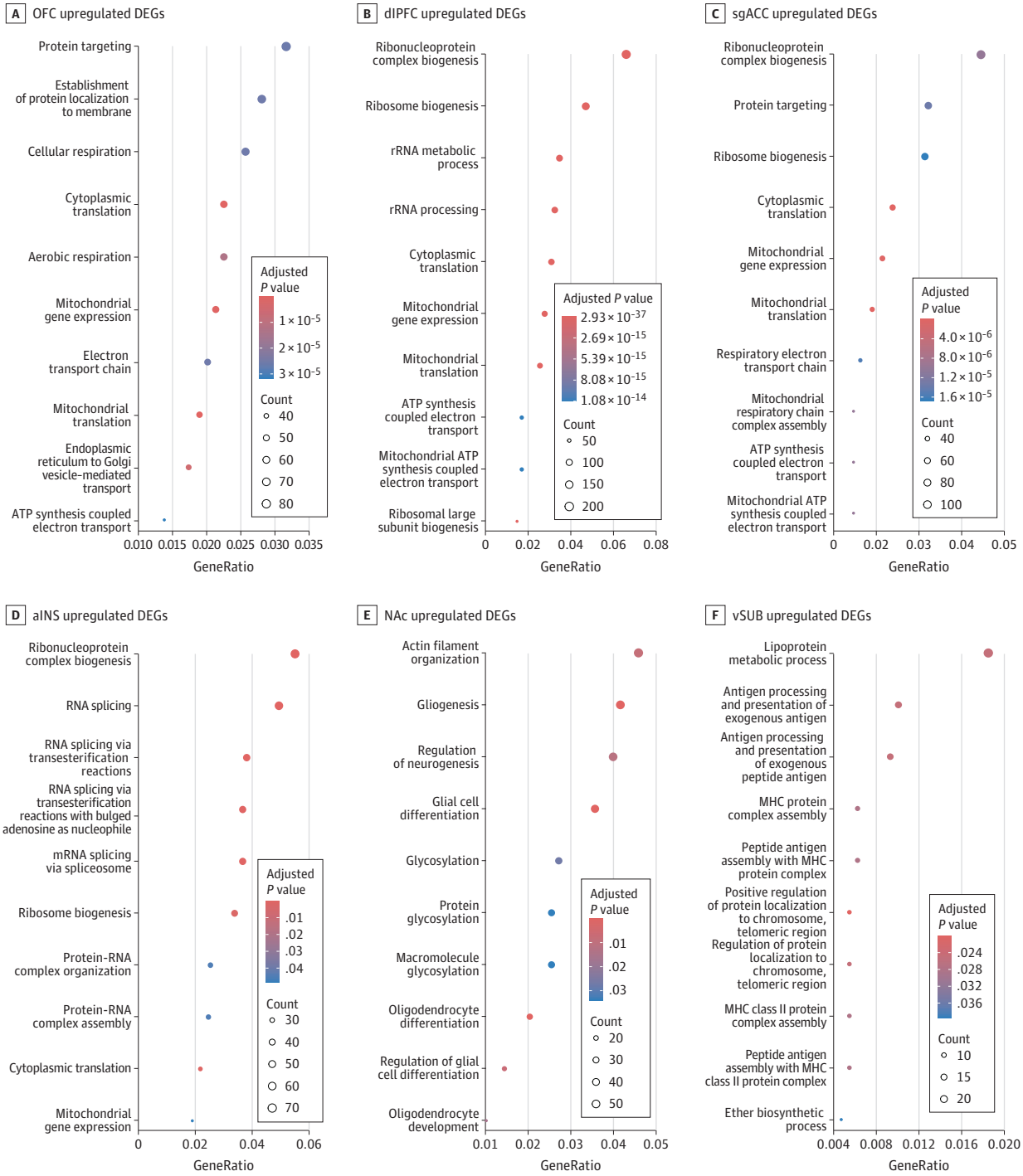
Figure 1. Brain Region-specific Molecular Pathways Upregulated in Major Depression



Gene ontology enrichment analysis was performed on upregulated differentially expressed genes (DEGs) from bulk RNA sequencing of postmortem brain tissue in individuals with major depressive disorder. Enrichment plots show the top 10 biological process terms per region and direction of regulation. A, In the orbitofrontal cortex (OFC), upregulated DEGs were involved in mitochondrial organization, protein targeting, and respiratory processes. B, In the dorsolateral prefrontal cortex (dlPFC), upregulated DEGs were associated with ribosome biogenesis, ribosomal RNA (rRNA) processing, and oxidative phosphorylation. C, In the subgenual anterior cingulate cortex (sgACC), upregulated DEGs showed enrichment for mitochondrial gene expression, respiratory chain assembly, and protein import. D, In the anterior insula (aINS), upregulated DEGs were linked to RNA splicing, ribonucleoprotein complex biogenesis, and translation. E, In the nucleus accumbens (NAc), upregulated DEGs were enriched for gliogenesis, regulation of neurogenesis, and glial cell differentiation. F, In the ventral subiculum (vSUB), upregulated DEGs were associated with antigen processing, major histocompatibility complex (MHC) assembly, and immune response. Midbody amygdala, temporal pole, dentate gyrus, and rostral anterior cingulate were excluded due to a lack of significant gene ontology terms.

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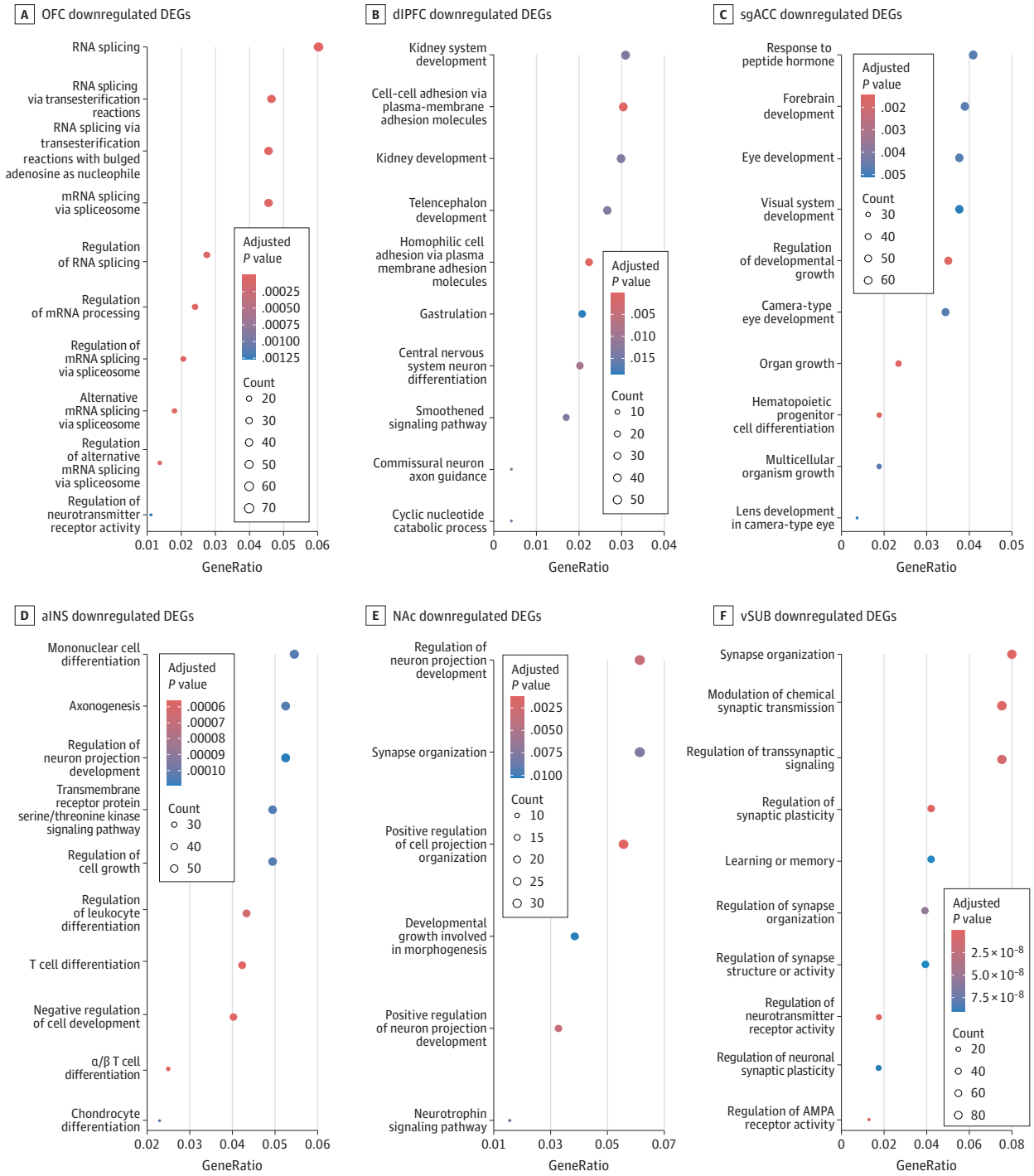
Figure 2. Brain Region-Specific Molecular Pathways Downregulated in Major Depression



Gene ontology enrichment analysis was performed on downregulated differentially expressed genes (DEGs) from bulk RNA sequencing of postmortem brain tissue in major depressive disorder. Enrichment plots show the top 10 biological process terms per region and direction of regulation. A, In the orbitofrontal cortex (OFC), downregulated DEGs were enriched for RNA splicing, messenger RNA (mRNA) processing, and neurotransmitter receptor activity. B, In the dorsolateral prefrontal cortex (dlPFC), downregulated DEGs were linked to nervous system development, kidney morphogenesis, and neuron differentiation. C, In the subgenual anterior cingulate cortex (sgACC), downregulated DEGs were enriched for visual system development, neurogenesis, and forebrain morphogenesis. D, In the anterior insula (aINS), downregulated DEGs were associated with T cell differentiation, axonogenesis, and regulation of cell growth. E, In the nucleus accumbens (NAc), downregulated DEGs were enriched for synapse organization, projection development, and neurotrophic signaling. F, In the ventral subiculum (vSUB), downregulated DEGs were linked to regulation of synaptic plasticity, learning and memory, and glutamate receptor signaling. Temporal pole, dentate gyrus, and rostral anterior cingulate excluded due to a lack of significant gene ontology terms.

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A large proportion of up- and downregulated DEGs across brain regions in MDD are RNA genes, including long noncoding RNAs (lncRNAs, eTable 2 in Supplement 2).

PPI network analysis (**Figure 3**A; eTable 12 in Supplement 2) revealed dlPFC hub genes implicated in DNA repair (*APEX1*) and protein transport (*ARL1*, *CCT7*, *ARFGEF1*). In the sgACC, hub gene *FBXO5* modulates DNA repair, *GMNN* contributes to DNA replication, *CDK6* to cellular development, *CORO1A* and *BLNK* are immune response-related, and *GPR65* regulates inflammation and apoptosis. OFC hub genes control synaptic transmission (*GLRA2*), immune response (*AHR*), cellular development (*E2F3*), and inhibitory signaling (*GABBR2*, *GABRB2*). In the aINS, hub gene *CAMK1G* regulates signal transduction, *MAP2K5* cellular development, *MAPK13* cellular stress, and *CSNK2A2* DNA repair. In the NAc, hub genes *DVL2*, *KITLG*, and *NANOG* are related to cellular development, and *HDAC10* to DNA repair. Amygdala hub genes included *CTCF* and *LGR5*, involved in cellular development, *TRAF4*, associated with apoptosis, and *TRIM8*, associated with immune function and apoptosis. In the vSUB, hub gene *ABCC9* is linked to ATP-sensitive potassium channel regulation, adjusting neuronal excitability in response to cellular energy, *ARHGEF6* to apoptosis, *GRIK5* and *CBLN2* to synaptic regulation, *IL7* to immune response, *ALDH2* to monoamine neurotransmitter synthesis, and *BMP6* to cellular development. The dentate gyrus had 2 hub genes: *GABBR1*, related to inhibitory signaling, and *ATF5*, involved in cellular development.

Whole Blood

The 12 blood transcriptomic studies[15,39-49] reported 472 blood DEGs in MDD, 349 upregulated, 105 downregulated, and 18 with conflicting findings across studies (eTable 3 in Supplement 2). Gene ontology analysis revealed upregulated translation, ribosome biogenesis, cellular respiration, oxidative phosphorylation, electron transport chain, and mitochondria function (eFigure 2A in Supplement 1). No downregulated gene ontology terms reached significance. PPI top hub genes (Figure 3B; eTable 13 in Supplement 2) included some coding for components of the 40S small ribosomal subunit (*RPL35*, *RPS11*, *RPS12*), essential for mRNA binding and translation, and others linked to mitochondrial ATP synthesis (*ATP5F1E*, *ATP5PO*) and immune response (*CCR5*).

Cell-Type Specific Transcriptomics

Brain Cell Types

The 6 brain snRNA-seq studies sharing DEG lists were conducted on the dlPFC,[49-53] and 1 study used laser capture microdissection to isolate sgACC cell types followed by RNA-seq.[32] Collectively they reported 4241 brain snRNA-seq DEGs in MDD (eTable 4 in Supplement 2).

Our gene ontology analysis (**Figure 4**) showed upregulated synaptic assembly and function, glutamate, and ion channel signaling in excitatory neurons, cell morphogenesis in inhibitory neurons, and cell-cell adhesion in both excitatory and inhibitory neurons. The sgACC study[32] reported upregulation of synapse assembly and migration in parvalbumin interneurons (eFigure 3 in Supplement 1). Astrocytes had upregulated process growth and nucleotide metabolism, and oligodendrocytes upregulated development, membrane transport, mitochondria function, electron transport, and calcium signaling (Figure 4).

Downregulated gene ontology terms in excitatory and inhibitory neurons were metabolism, synaptic vesicle signaling, cellular respiration, and ATP synthesis. Astrocytes had downregulated pathways that negatively regulate growth, and oligodendrocytes had downregulated gliogenesis and cell differentiation (Figure 4).

Our PPI analysis of dlPFC snRNA-seq DEGs identified cell-type specific hub genes (eFigure 4 in Supplement 1; eTable 14 in Supplement 2): excitatory neurons had *CAP2* and *ATXN1* linked to cellular development, *CAMK1G* involved in signal transduction, and *ATP2B3* and *CACNA1C* associated with calcium regulation; inhibitory neurons featured *RASD2* and *GLRB* (cell signaling), *CDKN1B* (development), and *PVALB* (inhibitory transmission). Microglia hub genes included proinflammatory *IL1B*, genes linked to cellular development (*XBP1*, *ASIC2*, *VWC2L*, *PCSK2*, *NEGR1*, and *GRID2*), *XKR4*, implicated in apoptosis, and glutamate receptor *GRIK3*. Astrocyte hub genes were associated with cell cycle regulation, apoptosis, and DNA repair (*GADD45B*), cell development (*DUSP1*, *APOE*) glutamate level regulation (*SLC1A3*), and GABA reuptake (*SLC6A11*).

Hub genes shared by snRNA-seq and bulk brain tissue transcriptomics (eFigure 5 in in Supplement 1; eTable 11 in Supplement 2) were involved in regulating cell morphology and migration (*ACTN4*), cell development (*CAPZB*), DNA damage repair (*BRD4*), ion homeostasis across cell membrane (*AQP4*, *ATP1A2*), and transcription (*EZR*).

Blood Cell Types

The 2748 blood cell RNA-seq DEGs were obtained sequencing presorted leukocytes[15], monocytes,[48] and peripheral blood mononuclear cells,[39,40,43] which include both monocytes and lymphocytes. These studies yielded 2661 DEGs in leukocytes, 30 in monocytes, and 57 in peripheral blood mononuclear cells (eTable 3 in Supplement 2).

Upregulated gene ontology pathways in leukocytes included RNA splicing, inflammation, and nuclear transport (eFigure 2B in Supplement 1). Other cell types did not have significant gene ontology terms. The same pathways were found upregulated combining all cell types (Figure 2) and when including data from whole blood reported above (eFigure 2D in Supplement 1). Like for whole blood, no downregulated gene ontology terms were found in any cell type.

PPI analysis of blood cell-type DEGs identified leukocytes hub genes linked to signal transduction (*ADCY3*) and cellular development (*CCND2*), while monocytes hub genes regulate transcription (*ZNF772*, *ZNF816*, *ZNF493*, *ZNF487*) and DNA repair (*TTI2*) (Figure 3; eTable 13 in Supplement 2).

Whole-Genome Methylation Analysis

Brain Methylomics

The 11 whole-genome methylation analysis studies included yielded 4207 brain DMGs (eTable 5 in Supplement 2) from the dlPFC (Brodman area 9),[14,19,28,54] anterior or frontopolar PFC (Brodman area 10),[55,56] PFC (subregion not specified),[57-59] sgACC,[60] OFC,[19,28,60] aINS,[19] NAc,[19] temporal pole,[34] and hippocampus (vSUB,[19] subfield not specified).[59]

Our gene ontology analysis revealed dlPFC hypomethylated genes linked to signal transduction (eFigure 6A in Supplement 1), while those in sgACC are involved in cell differentiation and stress response (eFigure 6B in Supplement 1). Hypermethylated dlPFC genes are involved in cell development, pattern specification (cell positioning, fate determination, spatial organization), neuron differentiation, synaptic vesicle transport, and RNA splicing (eFigure 6C in Supplement 1). No other brain region had significant gene ontology terms.

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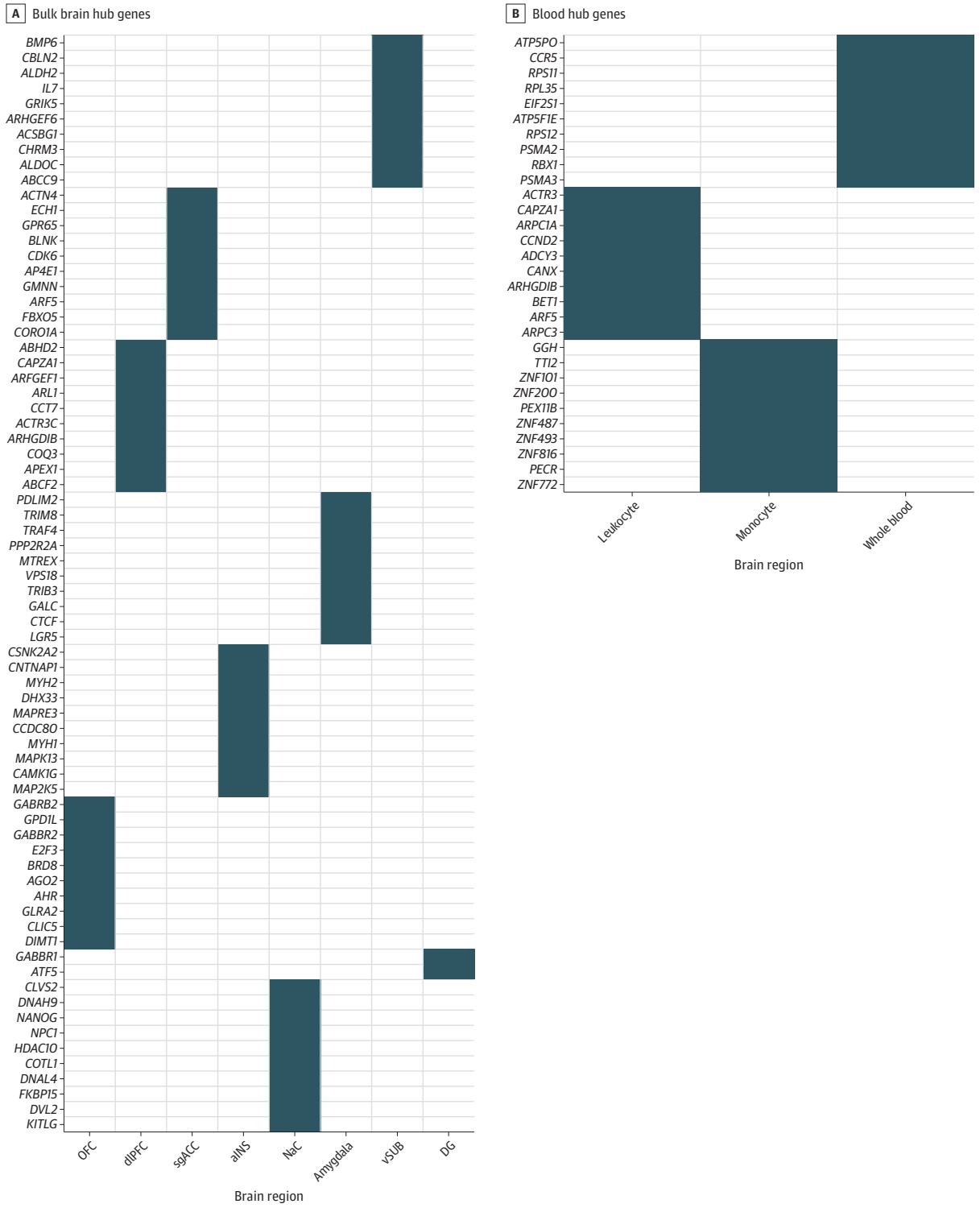
Figure 3. Heatmap of Brain and Blood Hub Genes Derived From Protein-Protein Interaction (PPI) Networks Constructed Using the Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) Database and Differentially Expressed Genes (DEGs) in Major Depressive Disorder (MDD)



A, Brain region–specific hub genes identified from bulk RNA sequencing data. Rows are hub genes identified from region-specific PPI networks using the top 300 protein-coding DEGs associated with MDD in each brain region. Columns correspond to the brain regions analyzed. Shaded cells indicate the presence of the hub gene within the STRING-derived PPI module for that region. The top 10 hub genes per brain region, ranked by node degree, are shown. B, Blood sample–specific hub genes identified from RNA sequencing data. Each row represents a hub gene identified from STRING-based PPI networks generated independently for each blood sample type using the top 300 protein-coding DEGs. Columns indicate type pf blood sample. Shaded cells indicate the presence of a hub gene within the PPI module for the blood sample type. The top 10 hub genes per blood group, ranked by node degree, are shown.

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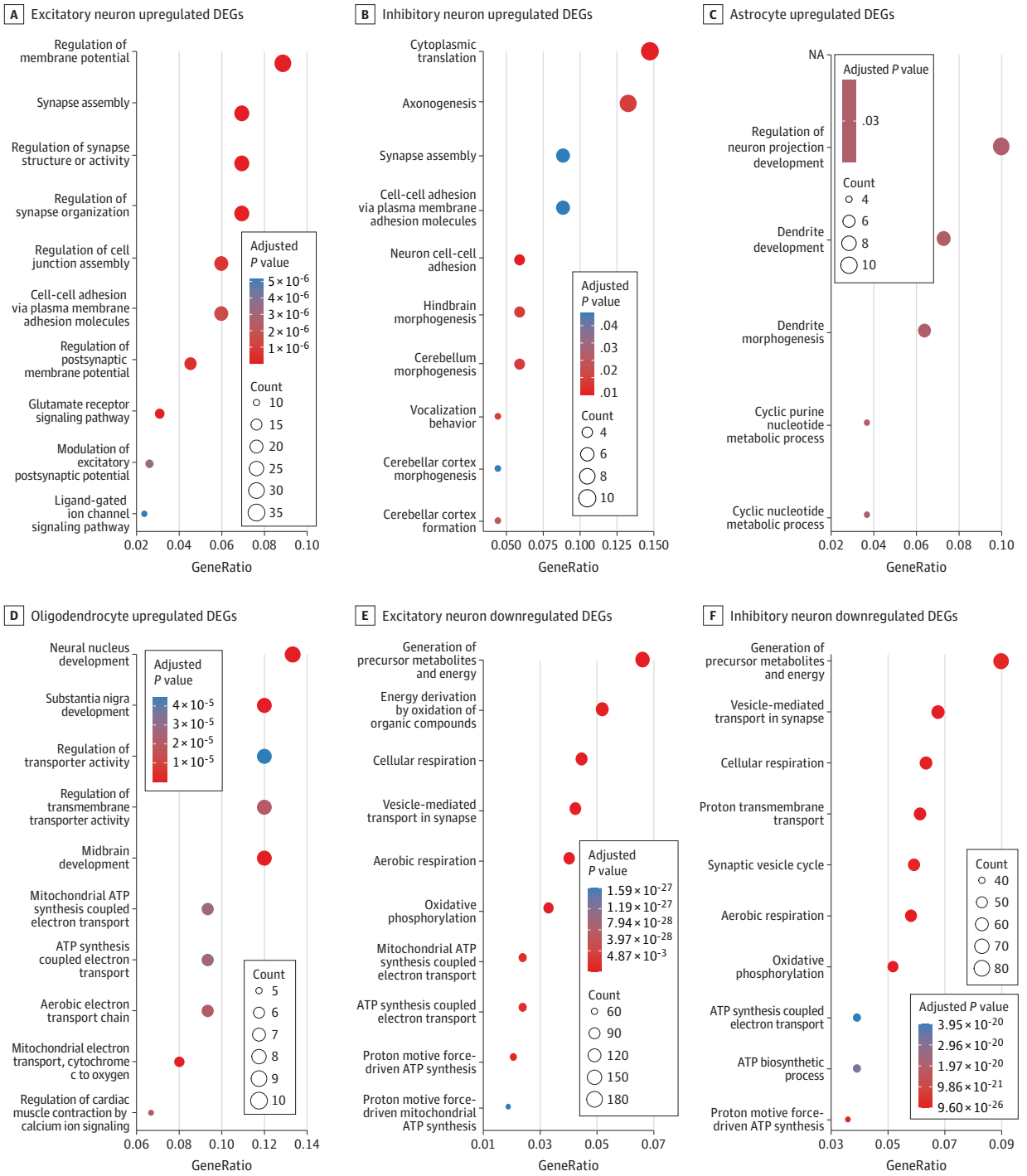
Figure 4. Dorsolateral Prefrontal Cortex Cell Type–Specific Molecular Pathways Altered in Major Depression



Gene ontology enrichment analysis (Biological Process category) was performed on upregulated and downregulated differentially expressed genes (DEGs) identified by single-nucleus RNA sequencing (snRNA-seq) across dlPFC cell types. Dot plots display the top ten enriched gene ontology terms per cell type and direction of regulation. A, In excitatory neurons, upregulated DEGs were enriched for regulation of membrane potential, synapse assembly, and glutamatergic signaling. B, In inhibitory neurons, upregulated DEGs were linked to axonogenesis, cell morphogenesis, and synaptic organization. C, In astrocytes, upregulated DEGs were associated with dendrite morphogenesis, cyclic nucleotide metabolism, and neuron projection development. D, In oligodendrocytes, upregulated DEGs were enriched for mitochondrial adenosine triphosphate (ATP) synthesis, oxidative phosphorylation, and protein translocation. E, In excitatory neurons, downregulated DEGs were enriched for mitochondrial energy production, oxidative phosphorylation, and synaptic vesicle-mediated transport. F, In inhibitory neurons, downregulated DEGs were linked to synaptic vesicle function, cellular respiration, and ATP synthesis. No significant gene ontology terms were identified in other cell types.

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Blood Methylomics

Fourteen whole-genome methylation analysis studies of whole blood,[47,55,56,61-69] white blood cells,[58] monocytes,[48] and deconvolution of T cells and monocytes[56] reported 8193 blood DMGs in MDD, 7464 in whole blood,[47,55,56,61-69] 3 in white blood cells,[58] 178 in monocytes,[48] 230 in T cells, and 318 in blood sample combinations (eTable 6 in Supplement 2).

Our gene ontology analysis showed that hypomethylated genes in all blood samples are involved in axonogenesis, cell differentiation, and signal transduction, (eFigure 7A in Supplement 1), similarly to the hypomethylated genes in whole blood, that also included those positively regulating nervous system development and axon guidance (eFigure 7B in Supplement 1), while only startle response genes were hypomethylated in T cells (eFigure 7C in Supplement 1).

Gene ontology terms associated with hypermethylated genes in combined blood cells and whole blood regulate cell developmental processes and morphogenesis (eFigure 7D and E in Supplement 1); hypermethylated genes in blood-derived cells regulate mitosis and NADPH regeneration (eFigure 7F and G in Supplement 1).

Blood and Brain Shared Transcriptional and Epigenetic Signatures: Candidate MDD Biomarkers

The blood and brain DEGs datasets shared 744 DEGs, 716 upregulated and 28 downregulated in both, while an additional 1507 DEGs had opposite dysregulation in MDD brain and blood (eTable 7 in Supplement 2). Among the 544 shared DMGs in blood and brain, 410 were hypomethylated and 134 hypermethylated in both. An additional 1076 had opposite methylation patterns in blood and brain, and for 49 reported DMGs, whether they were hyper- or hypomethylated was not described (eTable 8 in Supplement 2).

The 716 upregulated DEGs in brain and blood regulate apoptosis (*FAM32A*, *BCLAF1*, *PDCD6*, *GOS2*, *SRGN*), autophagy (*DRAM1*, *WDR45*, *BECN1*, *UBQLN2*, *LGALS8*), immune responses (*FCGR1A*, *LACC1*, *PADI4*, *PARP14*, *MORC3*), and signal transduction, including GPCR-mediated calcium signaling, PI3K- and Rho GTPase–dependent cytoskeletal regulation, redox-sensitive ion channel activity, and immune receptor signaling (*PLCB1*, *MYO9A*, *CLIC1*, *ARAP3*, *CD53*).

The 410 hypomethylated genes in brain and blood regulate cell division and differentiation (*ZBTB1*, *FSTL5*, *RAI14*, *MPLKIP*), immune responses (*FASLG*, *ANKRD17*, *TRIM10*), transcriptional regulation (*SMARCA4*, *ARNT2*, *CREB3L2*, *ZNF215*, *ZNF317*), and G protein–coupled receptor signaling (*VIPR1*, *NPSR1*, *ARHGEF1*, *OR2H1*, *OR6N1*).

The 28 downregulated DEGs in brain and blood control morphogenesis and development (*PTCH1*, *CROCC*, *NPTX1*), G protein–coupled receptor signaling (*CELSR2*, *NPFFR1*), and signal transduction (*IL1RAP*, *CSPG4*).

The 134 hypermethylated genes in brain and blood control apoptotic processes (*BLCAP*, *CADM1*, *BCL7C*, *EMP2*, *BAD*), cell differentiation (*FEZF2*, *PDLIM7*, *NDRG4*, *SPATA24*, *ARMC2*), and immune responses (*BCL6*, *DEFB124*, *ECSIT*).

Hub gene analysis of both blood and brain DEG PPI networks pointed to genes controlling cellular development (*CAPZB*) and immune response (*AIF1*, *B2M*, *CD44*, *CASP1*) (eFigure 8 in in Supplement 1; eTable 11 in Supplement 2).

Integrating Transcriptional and Epigenetic Signatures With GWAS Genetic Variants

We found that 132 of the blood and brain DEGs were MDD hits in the NHGRI-EBI GWAS Catalog[17] (eTable 9 in Supplement 2), of which 39 had RNA overexpression, 4 were downregulated, and 89 had RNA regulation in opposite directions in blood and brain. These include genes regulating neurodevelopment (*TCF4*, *MDGA1*), immune response (*LY86*), and apoptosis (*DFFA*, *CASP8AP2*).

DMGs in MDD brain and blood corresponded to 158 GWAS hits (eTable 10 in Supplement 2), and included other genes linked to neurodevelopment (*RTN1*, *NRG1*), immune response (*COLEC11*, *TRIM26*), and apoptosis (*BLCAP*, *BCL7C*, *CADM1*, *DCC*).

Discussion

Multiomic profiling of cellular dysregulation contributing to MDD pathogenesis remains limited, and we currently lack molecularly guided precision medicine treatments and biomarkers. This systematic review integrating brain and blood DEG and DMG lists from 54 bulk and cell type–specific transcriptomic and methylomic studies and GWAS databases, highlights converging dysregulated molecular pathways in MDD. Disruptions in mitochondrial metabolism, immune signaling, transcriptional control, neurodevelopment, apoptosis, and DNA repair appear central to the pathophysiology of MDD (**Figure 5**).

Mitochondrial function upregulation across dlPFC, sgACC, and OFC[35,57,60,70] tissue and in dlPFC oligodendrocytes suggests increased oxidative stress in cells supporting axon function and neurotransmission. Meanwhile, dlPFC, sgACC, and OFC excitatory and inhibitory neurons had decreased energy, possibly affecting executive functions,[3] decision-making,[4] and impulse control[5] in MDD. Because blood DEGs were also enriched in mitochondrial pathways, those genes may become MDD biomarkers useful for patient stratification.

Neurodevelopmental pathway downregulation in bulk dlPFC, sgACC, aINS, and NAc, including axonogenesis and projection development, suggests impaired structural plasticity in regions critical for executive functions[3]; decision-making[4]; integration of sensory, emotional, cognitive, and interoceptive information[11]; and reward.[10] Supporting the central role of neurodevelopmental functions in MDD were hub genes found not only in the dlPFC; sgACC; aINS; NAc; and dlPFC excitatory and inhibitory neurons, microglia, and astrocytes, but also in the amygdala, hippocampus vSUB, and dentate gyrus, which control memory and emotional responses.[6] Interestingly, other developmental pathways' upregulation in dlPFC and sgACC tissue and in dlPFC excitatory and inhibitory neurons, astrocytes, and oligodendrocytes, suggests brain reorganization and region-specific changes in cellular plasticity, possibly explaining altered functional connectivity of these regions in MDD.[4,10,11,71] Upregulated gliogenesis in the NAc, a region involved in reward,[10] suggests a role of glia in shaping connectivity and function in this region, which has altered resting-state functional connectivity with the PFC in MDD, which appears reversable with cognitive-behavioral therapy.[10] Hypomethylated axonogenesis genes in blood and neurodevelopment-related GWAS hits overlapping with brain DEGs (*TCF4*, *MDGA1*) and DMGs (*RTN1*, *NRG1*) may serve as useful biomarkers for MDD stratification. It should be noted that without

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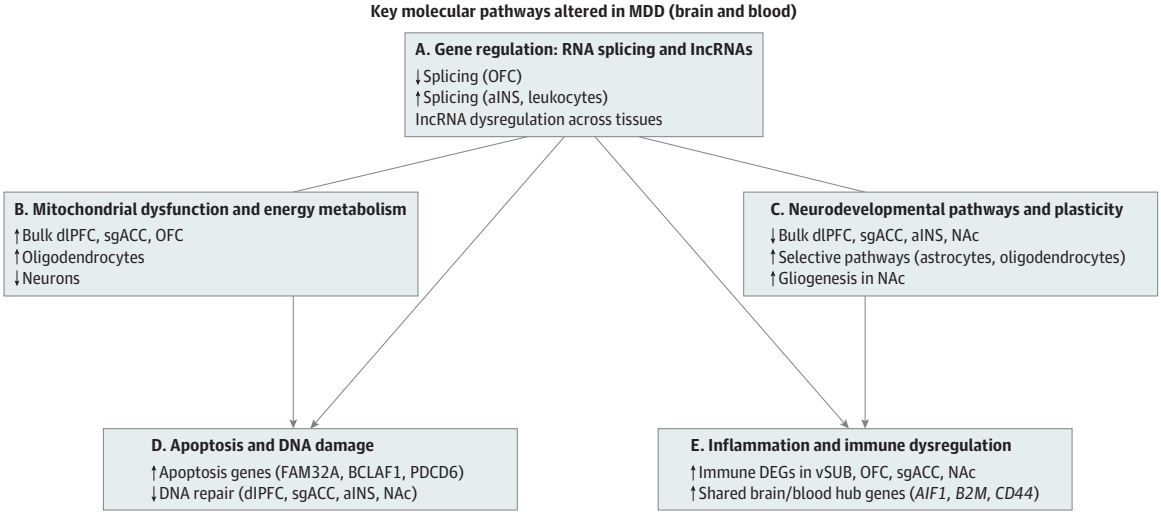
Figure 5. Molecular Pathways Suggested to Contribute to Brain Cellular Damage in Major Depressive Disorder (MDD)



A, RNA splicing was downregulated in the orbitofrontal cortex (OFC), upregulated in the anterior insula (aINS) and blood, and hypermethylated in the dorsolateral prefrontal cortex (dlPFC), suggesting ongoing epigenetic and transcriptional regulation affecting gene expression in MDD. B, Upregulated mitochondria genes in bulk dlPFC, subgenual anterior cingulate cortex (sgACC), OFC, and dlPFC oligodendrocytes suggest increased oxidative stress. Downregulated mitochondria genes suggest decreased cellular respiration and energy deficits in excitatory and inhibitory neurons, likely contributing to impaired executive functions, decision-making, and impulse control. Mitochondrial DEGs have the potential to become biomarkers of major depressive disorder, given they are also found dysregulated in patient blood. C, Axonogenesis and neuron projection development were downregulated in dlPFC, sgACC, aINS, and nucleus accumbens (NAc). Hub genes for neurodevelopment were found in neurons, astrocytes, microglia, oligodendrocytes, amygdala, hippocampus ventral subiculum (vSUB), and dentate gyrus (DG). Gliogenesis was selectively upregulated in NAc, and there were upregulated developmental genes in specific cell types of the dlPFC and sgACC. Identified blood biomarkers were hypomethylated axonogenesis genes, and genome-wide association study (GWAS) gene variants (*TCF4*, *MDGA1*, *RTN1*, *NRG1*). D, Upregulated apoptosis genes included *FAM32A*, *BCLAF1*, *PDCD6*, *GOS2*, and *SRGN*. Apoptosis hub genes were found in sgACC, amygdala, dlPFC microglia, and astrocytes. There was overlap between GWAS hits, DEGs, and differentially methylated regions for the following genes: *DFFA*, *CASP8AP2*, *BLCAP*, *BCL7C*, *CADM1*, and *DCC*. Downregulated genes were related to DNA repair in the amygdala, dlPFC, sgACC, aINS, NAc, and monocytes. These findings indicating neuronal or glial apoptosis explain the findings of reduced PFC and hippocampus volume in MDD. E, Upregulated immune DEGs were found in vSUB, OFC, sgACC, NAc, and blood. Shared hub genes included *AIF1*, *B2M*, and *CD44*. There was GWAS overlap with the following genes: *LY86*, *COLEC11*, and *TRIM26*. The findings suggest central and peripheral immune dysfunction, which could be a biomarker for potential stratification of MDD subtypes.

longitudinal evidence, the pathogenetic meaning of these findings and relative blood biomarkers remains to be determined.

Apoptosis-related gene upregulation in brain and blood (*FAM32A*, *BCLAF1*, *PDCD6*, *GOS2*, *SRGN*); apoptosis hub genes in sgACC and amygdala tissue and dlPFC microglia and astrocytes; and GWAS hits overlapping with DEGs (*DFFA*, *CASP8AP2*) and DMGs (*BLCAP*, *BCL7C*, *CADM1*, *DCC*) may explain reduced PFC and hippocampus volume in MDD.[3,5,7] DNA damage may contribute to apoptosis, as DNA repair genes were downregulated in amygdala and were hub genes in the dlPFC sgACC, aINS, NAc, and blood monocytes. Cellular stress and increased apoptosis likely contribute to brain atrophy and altered circuit connectivity in MDD. DNA repair genes altered in blood, if confirmed, may provide markers for a possible MDD molecular endophenotype. Functional studies would be needed to determine whether apoptosis is causal or a downstream cellular stress response in MDD.

Inflammatory pathway upregulation in MDD is evidenced by vSUB DEGs; neuroinflammation hub genes in OFC, sgACC, NAc, and vSUB; overlapping brain and blood DEGs (*FCGR1A*, *LACC1*, *PADI4*, *PARP14*, *MORC3*); converging blood and brain hub gens (*AIF1*, *B2M*, *CD44*); and GWAS hits[17] corresponding to DEGs (*LY86*) or DMGs (*COLEC11*, *TRIM26*). This confirms that central and peripheral immune dysfunction is critical to MDD pathophysiology[44] and may provide useful to distinguish patients in which neuroinflammation may be the disease driver. The cross-sectional nature of the studies does not allow determining whether inflammation is a cause or a consequence or a transient state or a stable marker of MDD.

RNA splicing downregulation in the OFC and upregulation in the aINS and leucocytes, together with hypermethylation of RNA splicing genes in dlPFC, likely affect gene expression, suggesting regulatory mechanisms possibly driven by epigenetic factors. RNA splicing and lncRNA regulation is a primary step in gene expression and could precede and drive dysfunction in other pathways. If splicing is dysregulated, abnormal isoforms of mitochondrial proteins, apoptosis regulators, neurodevelopmental genes, or immune mediators are obtained. On the other hand, oxidative stress, inflammation, or DNA damage can impair the spliceosome, leading to secondary changes in splicing. For example, inflammatory signaling can alter RNA-binding proteins, changing downstream splicing. These findings of disrupted transcriptional regulation in MDD support previous studies in depressed suicides.[29]

Opposite methylation changes in brain and blood may reflect concerted tissue-specific epigenetic regulation. Blood DNA methylation reflects mainly immune cell populations, while brain methylation is driven by neurons, astrocytes, oligodendrocytes, and microglia. As cortisol and inflammation change concurrently with stress,[72] a stress-response gene could be silenced in the brain (reducing excitability or plasticity) and activated in blood immune cells

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(to mount inflammatory or stress-related responses). With chronic low-grade systemic inflammation,[43] cytokines signal to the brain, where epigenetic programs may suppress neural plasticity genes to prevent excitotoxicity in MDD. Moreover, DNA methylation can have distinct effects, promoter hypermethylation usually represses transcription, while gene body methylation can increase expression. Therefore, if the methylation changes across tissues occur in different genomic contexts (eg, promoter vs enhancer vs gene body), the apparent opposite directionality of methylation across tissues may lead to the same result. Additionally, DMGs may show mirror regulation across tissues, as neurotrophic genes suppressed in the brain (impairing neurogenesis and neuroplasticity) may get upregulated in peripheral tissues to maintain systemic levels of neurotrophic factors (eg, brain-derived neurotrophic factor). Therefore, apparent opposite methylation directions could reflect coordinated responses between peripheral and brain cell types.

Sex-specific molecular alterations, including RNA regulation,[20-23,52] suggest divergent pathogenesis between men and women, which requires further studies. Variability in patient medication status, clinical endophenotypes, comorbidities, demographic composition, and technical platforms likely contributes to the heterogeneity of results across studies. The extent to which the observed epigenetic and transcriptional signatures reflect genetic or environmental causal mechanisms vs downstream consequences of the disease remains unresolved.

Many of the dysregulated transcriptomic signatures identified in MDD appear transdiagnostic and are shared with schizophrenia, bipolar disorder, autism, and Alzheimer disease.[73,74] This may raise concerns about disease specificity or rather offer an opportunity to revisit a diagnostic system that is based on syndromic presentations instead of molecular pathogenetic pathways. We could hypothesize that the same molecular disruption may result in different symptoms depending on the brain region affected, similarly to what happens with a stroke. Detailed metadata reporting and transdiagnostic single-cell multiomic studies are warranted to detect biology-based clinical phenotypes in psychiatry. Longitudinal and prospective studies will be critical for testing biomarkers by integrating molecular, imaging, and clinical features.

Emerging spatial transcriptomics technologies, allowing profiling gene expression and cell functional states in the anatomical landscape, could be used to map DEGs and DMGs for a deeper understanding of their functional significance in the brain circuit. Moreover, spatial differential gene expression and chromatin accessibility analysis will allow detecting genetic alterations that are brain layer specific.[75] Additionally, spatial approaches allow investigating transcripts in cytoplasm and processes rather than only in nuclei. Integration of single-nucleus RNA and chromatin accessibility sequencing with proteomics, methylomics, and genetic variation in the same subjects, will offer insights on transcriptomic and epigenetic changes that ultimately result in protein expression alterations.

Limitations

Limitations of this work include patient and methodologies heterogeneity across studies, which likely limit findings to altered molecular pathways that are shared by all MDD phenotypes, and not reflecting MDD subtypes. Similarly, methodological dissimilarities, and differences in baseline gene expression between brain and blood narrow overlapping statistical significance to only the most powerful and widespread DEGs and DMGs that, despite the variability in sequencing platforms, normalization pipelines, and statistical thresholds, emerge in multiple studies. Improved inclusion of diverse ancestries and larger samples will enhance the generalizability of findings and reduce bias in future biomarker strategies.

Conclusion

In summary, convergent biological processes disrupted in MDD brain and blood include transcriptional regulation, mitochondrial metabolism, neurodevelopment, inflammation, and apoptosis. These may result from a shared pathogenetic cascade or represent molecular endophenotypes of the disease. Continued integration of multiomic and clinical data will be essential to move beyond symptom-based disease classification toward biologically informed precision medicine approaches for diagnosing and treating psychiatric illness.

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